# SDLC Using Agentic AI

Building an Adaptive AI Software Engineering Organization

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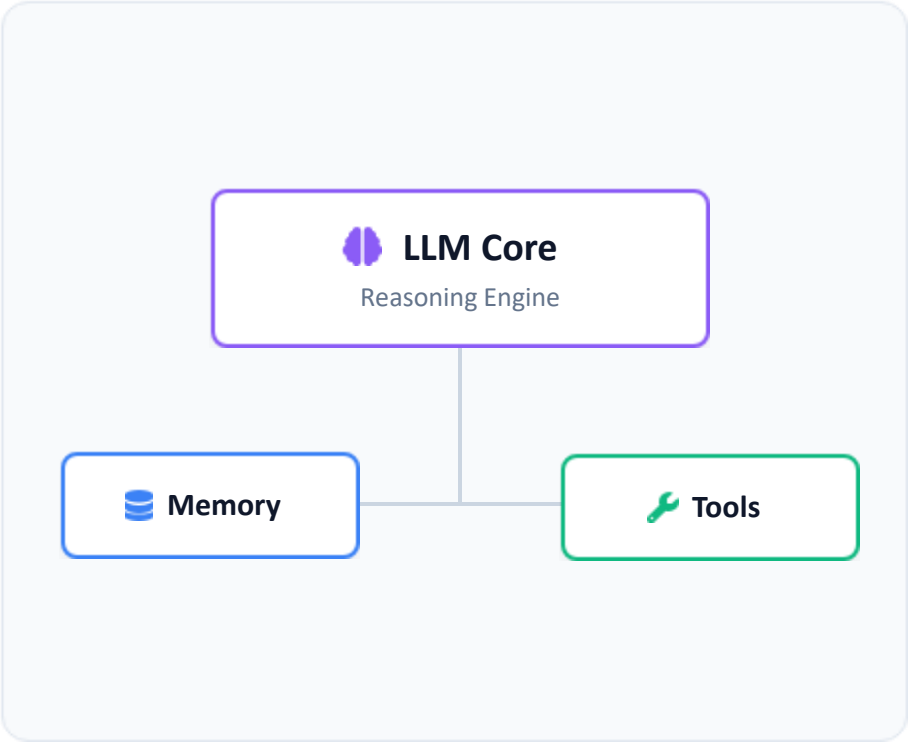
# What is an AI Agent?

An **AI Agent** is an autonomous system that uses a Large Language Model (LLM) as its central reasoning engine to actively solve problems.

It observes its environment, breaks down complex goals into actionable steps, and manipulates its surroundings by calling external tools.

**The Formula:**

```
Agent = LLM + Memory + Planning + Tools
```



# Types of AI Agents

## Simple Reflex

React solely to current percepts without considering past experiences, following predefined condition–action rules.

- No memory capacity
- Rule-based decisions
- Fast but inflexible

EXAMPLES

Auto doors, Thermostats, Motion lights

## Model-Based

Maintain an internal model (state) of the environment, allowing informed decisions despite partial observability.

- Maintains internal state
- Handles partial observability
- Updates knowledge continuously

EXAMPLES

Robot vacuums, Warehouse robots

## Goal-Based

Choose actions by evaluating future states and planning a sequence of steps to achieve a specific objective.

- Goal-oriented approach
- Planning & search capabilities
- Evaluates possible actions

EXAMPLES

GPS navigation, Logistics planners

## Utility-Based

Evaluate multiple possible actions using a utility function, selecting the option that maximizes overall performance and handles trade-offs efficiently.

- Decision optimization
- Handles complex trade-offs
- Maximizes expected reward/utility

EXAMPLES

Algorithmic trading, Dynamic pricing, Autonomous driving

## Learning Agents

Improve behavior autonomously over time by continuously learning from data, user feedback, and environment interactions.

- Self-improving over time
- Learns directly from experience
- Adapts to changing environments

EXAMPLES

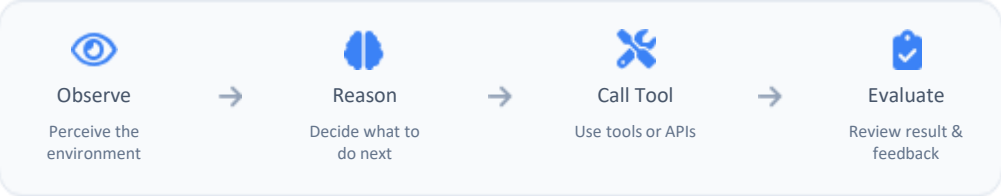
AlphaGo, Netflix Recommendations, Adaptive spam filters

# AI Agent Architectures

Different patterns for designing agents based on how they reason, act, and collaborate.

## 1. ReAct (Reason & Act)

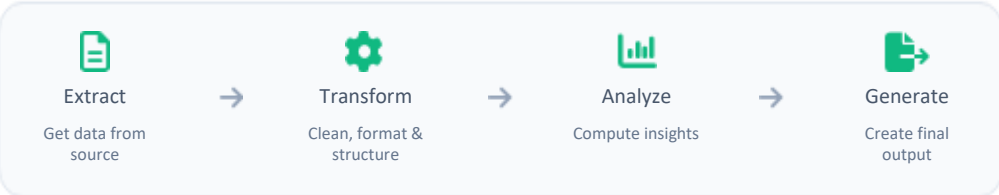
A single agent follows an iterative loop: observe the environment, reason about what to do next, act using tools, evaluate the result, and repeat until the task is complete.




Example Use Cases: Q&A, Code generation & debugging

## 2. Sequential Workflow

A pipeline where tasks are executed in a fixed, linear order. The output of one step becomes the input to the next.



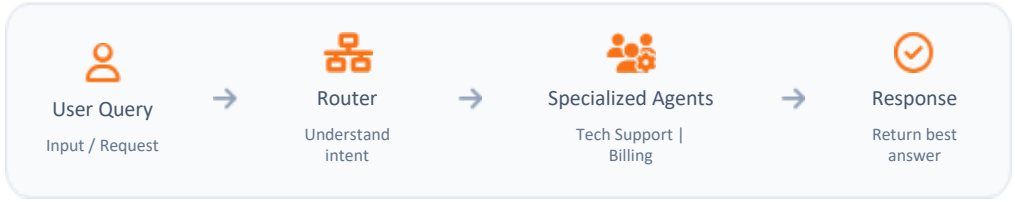

Example Use Cases: Document processing pipeline, Report generation

# AI Agent Architectures

Different patterns for designing agents based on how they reason, act, and collaborate.

## 3. Routing Architecture

A supervisor (router) interprets the intent of the prompt and routes it to the most suitable specialized worker agent.




Example Use Cases: Customer support automation

## 4. Multi-Agent Swarm

Multiple agents with different roles collaborate, communicate, and debate to solve complex problems under a coordinating agent.


Example Use Cases: Software development teams, Threat analysis & support

# The Traditional SDLC

The Traditional Software Development Life Cycle (SDLC) is the structured framework that engineering teams follow to plan, design, build, test, and deploy software applications from initial concept.


This Manual SDLC execution creates severe friction—engineers waste most of their time on administrative handoffs, environment setups, regression testing, and ticketing overhead instead of writing core logic.

**So, Our Aim is to automate the Traditional SDLC with Agentic AI,** which will replaces these static bottlenecks with continuous, self-healing pipelines that autonomously execute tests, patch build failures, flag vulnerabilities, and manage releases. This **eliminates human error, compresses cycle times from months to hours, and frees development teams to focus on high-value architecture and innovation.**

# Beyond Code Generation: Agentic AI

Agentic AI consists of autonomous AI agents capable of planning, reasoning, using tools, maintaining memory, and collaborating to achieve complex goals.

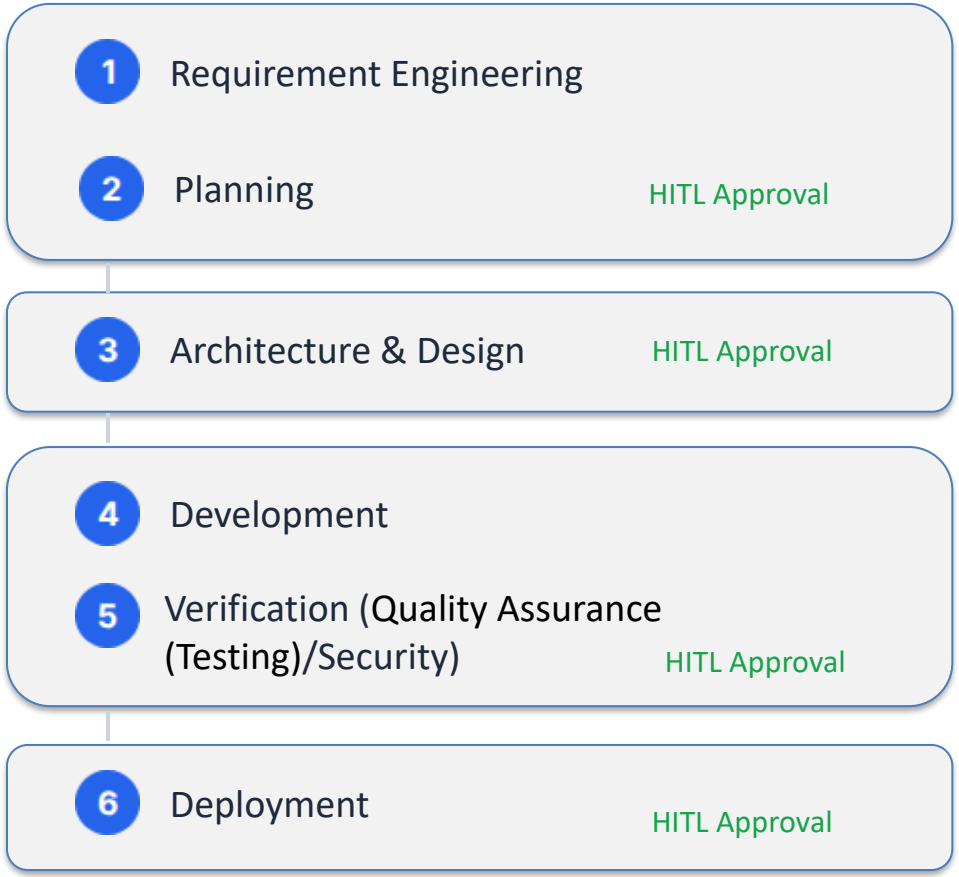
| Capability | Standard Chatbot | Agentic AI |
| --- | --- | --- |
| **Execution** | Passive (Answers queries) | Autonomous (Acts & Uses Tools) |
| **Workflow** | Single-step generation | Multi-step reasoning & execution |
| **Context** | No Memory (Session bound) | Persistent Memory (Context aware) |
| **Strategy** | No Planning | Goal Planning & Reflection |

# Agentic Frameworks

| Framework | Execution Model | Core Strengths | Key Limitations | Ideal Use Cases | Offline / Local Capability |
|---|---|---|---|---|---|
| **LangChain** | Chained Prompting & Tools | Rich toolset, extensive RAG ecosystem, vast API integrations | Workflow complexity at scale, abstractions can hide errors | General LLM applications, retrieval-augmented generation | **Excellent:** Natively connects to local LLMs (Ollama, Llama.cpp) and local vector stores (Chroma). |
| **LangGraph** | Stateful Graph Orchestration | Cycles and loops, robust state persistence, time-travel debugging | Steeper learning curve, excessive setup for simple tasks | Production-grade multi-agent systems, human-in-the-loop control | **Excellent:** Inherits LangChain's local capabilities; operates entirely offline with self-hosted models. |
| **CrewAI** | Role-based Team Collaboration | Easy setup, intuitive agent personas, automatic task delegation | Limited dynamic branching and complex workflow flow control | Autonomous research pipelines, content creation, structured teams | **Strong:** Explicitly supports local execution by binding agents to local endpoints via Ollama or LiteLLM. |
| **OpenAI Agents SDK** | Lightweight Handoffs & Guardrails | Minimal abstractions, built-in tracing, standardized routing primitives | Lacks built-in persistent long-term memory or native graph structures | Rapid deployment of production agents, voice pipelines, clear delegation | **Very Good:** Provider-agnostic design allows it to run offline by routing to local models via the Chat Completions API. |

# Proposed Architecture: Modular Pipeline

A modular pipeline with integrated **Human-in-the-Loop (HITL)** governance gates to ensure enterprise safety.

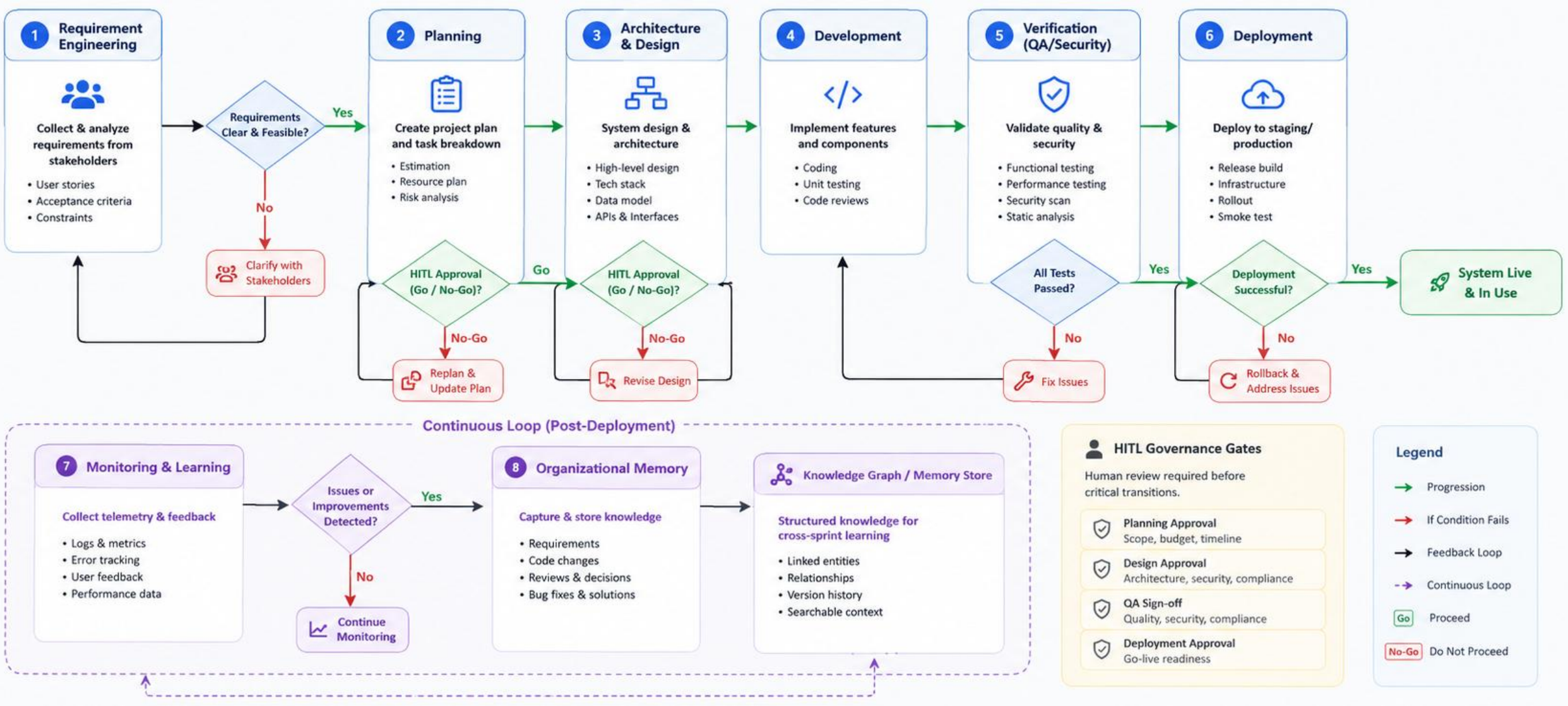



## Continuous Loop

**7. Monitoring & Learning:** Consumes production logs, metrics, and user feedback to detect errors and optimize prompts.

**8. Organizational Memory:** Captures all requirements, code, reviews, and bug fixes into a Knowledge Graph for cross-sprint learning.

# Proposed Architecture: Modular Pipeline



# Identifying the Gap

While highly capable, existing autonomous systems index heavily on code generation, lacking end-to-end SDLC orchestration and organizational memory.

| System | Requirements | Design | Coding | Testing | Deployment |
|---|---|---|---|---|---|
| **Devin / OpenHands** | Partial | Partial | Yes | Yes | Partial |
| **MetaGPT / ChatDev** | Yes | Yes | Yes | Partial | No |
| **SWE-Agent** | No | No | Yes | Yes | No |
| **Proposed System** | Yes | Yes | Yes | Yes | Yes |

# Core Innovation: Adaptive Team Generation

## Traditional: Fixed Agents

Standard agentic systems rely on rigid, hardcoded workflows (e.g., always Planner → Developer → Tester).

- Inefficient for small tasks (over-engineered).
- Lacks specialized roles (Security, DevOps) for complex enterprise apps.

## Proposed: Dynamic Generation

The Orchestrator analyzes project complexity, risk, and size before execution to spawn agents on-the-fly.

- **Small Script:** Spawns Planner + Developer.
- **Enterprise App:** Spawns Architect, Backend, Frontend, Security, QA, DevOps.
- Agents are retired when tasks complete, saving compute.

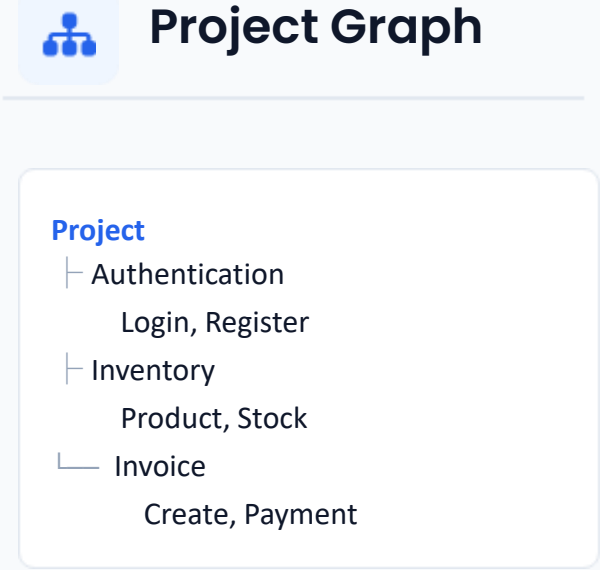
# Core Innovation: Hierarchical Software Knowledge Graph (HSKG)

Beyond Traditional Memory:
A Context-Aware Project Memory

**The Problem:** Current AI agents retrieve entire repositories or large documents, resulting in high token consumption and irrelevant context.

**Our Solution:** Our proposed HSKG stores the software project as a hierarchical graph, enabling agents to retrieve only the smallest relevant subgraph required for a task.

## Project Graph




**Purpose**

- Represents the complete SDLC structure
- Maintains hierarchy between project components
- Enables modular navigation

## Context-Aware

**Instead of:**
Entire Repository → LLM

**Agent retrieves:**

Login Module -> UI -> Backend Script & Variables -> API -> User DB

**Benefits**

- Minimal context
- Lower token usage
- Faster reasoning
- Better accuracy
- Reduced hallucination

# Local Offline Execution Strategy

Deploying via offline models ensures data privacy. Model selection is based on VRAM footprint, Training data freshness, Suitable for Agentic Operations.

| Model (4-bit Quant) | Recommended Hardware (VRAM) | Knowledge Cutoff | Agent Capability | Best Use Case |
|---|---|---|---|---|
| **Gemma 4 4B** | Laptop / iGPU (~5 GB) | Early 2026 | ⭐⭐⭐ | Basic assistants, RAG, lightweight offline agents |
| **Gemma 4 12B** | RTX 4070 (8–10 GB) | Mid 2026 | ⭐⭐⭐⭐ | General-purpose AI agents with good reasoning & tool use |
| **Mistral Small 3.2 (24B)** | RTX 4090 (16 GB) | Mid 2025 | ⭐⭐⭐⭐⭐ | Enterprise agents, multi-tool workflows, planning |
| **Qwen3-Coder 32B** | Workstation / Mac M-Series (20–22 GB) | Mid 2025 | ⭐⭐⭐⭐⭐ | Coding agents, debugging, software engineering |
| **Llama 3.3 70B** | Multi-GPU / High-End Server (48+ GB) | Late 2023* | ⭐⭐⭐⭐⭐ | Research, complex planning, multi-agent orchestration |

# Conclusion

We are proposing an **Autonomous Software Development Lifecycle (A-SDLC)** framework that transforms traditional software engineering into an **AI-Orchestrated Software Engineering Organization**.

Instead of isolated coding assistants, the system coordinates multiple specialized AI agents across the complete SDLC—from **requirements engineering** to **continuous monitoring and learning**.

## Expected Outcomes

- Reduced context size and token consumption
- Faster and more accurate software development
- Improved traceability across the SDLC
- Persistent engineering knowledge

# Thank You